

FOOD WASTE MANAGEMENT SYSTEM

1. Business Objective :

To reduce food wastage by connecting food providers (restaurants, grocery stores, supermarkets, etc.) with receivers (NGOs, community centers, needy individuals) through a centralized platform.

Simple Explanation

Many restaurants throw away extra food every day.

At the same time, many people struggle to get food.

This system helps:

- Food Providers donate extra food
 - NGOs/People claim food
 - Food gets distributed instead of wasted
 - Food wastage decreases
 - Hunger issues are reduced
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2. Business Problem :

Current Problems

1. Excess food is wasted.
2. NGOs don't know where food is available.
3. No centralized platform exists.
4. Food distribution is inefficient.
5. Providers and receivers cannot easily connect.
6. No proper analysis of food donation trends.

Solution

Create a system where:

- Providers list food
- Receivers claim food
- Claims are tracked

- Data is stored in SQL
- Analysis is performed
- Streamlit app provides an interface

3. Dataset Understanding :

The dataset underlying the Food Waste Management System consists of four interrelated tables. They are as follows:

Table Name	Purpose
providers_data	Stores master information about every registered food provider, including contact details and location.
receivers_data	Stores master information about every registered food receiver, including contact details and location.
food_listings_data	Stores details of individual food items listed by providers, including quantity, type, and expiry information.
claims_data	Stores transactional records of claims made by receivers against specific food listings, along with claim status and timestamp.

3.1 providers_data :

Field	Description
provider_id	Unique identifier assigned to each food provider; serves as the primary key of the table.
name	Registered name of the provider, such as the restaurant, store, or caterer name.
type	Category of the provider, for example restaurant, grocery store, or caterer.
address	Physical address at which the provider is located.
city	City in which the provider operates, used extensively for geographic analysis.
contact	Contact information, such as a phone number, used to reach the provider.

3.2 receivers_data :

Field	Description
receiver_id	Unique identifier assigned to each food receiver; serves as the primary key of the table.
name	Registered name of the receiver, such as an individual, NGO, or community centre.
type	Category of the receiver, for example individual or organisation.
city	City in which the receiver is located, used for matching with nearby providers.
contact	Contact information used to communicate with the receiver regarding claims.

3.3 food_listing_data :

Field	Description
food_id	Unique identifier assigned to each food listing; serves as the primary key of the table.
food_name	Descriptive name of the food item being listed, such as rice, bread, or curry.
quantity	Numeric quantity of the food item available, typically expressed in servings or units.
expiry_date	Date by which the listed food item should be collected and consumed.
provider_id	Foreign key referencing the provider in providers_data who created the listing.
provider_type	Category of the listing provider, denormalised here to simplify certain analytical queries.
location	Specific location or city associated with the listing, used for receiver-side filtering.
food_type	Dietary classification of the food, such as vegetarian, non-vegetarian, or vegan.
meal_type	Meal category of the food, such as breakfast, lunch, dinner, or snacks.

3.4 claims_data :

Field	Description
claim_id	Unique identifier assigned to each claim transaction; serves as the primary key of the table.
food_id	Foreign key referencing the specific food listing being claimed in food_listings_data.
receiver_id	Foreign key referencing the receiver in receivers_data who initiated the claim.
status	Current state of the claim, recorded as Completed, Pending, or Cancelled.
timestamp	Date and time at which the claim was recorded; the source field from which claim_date, claim_time, and weekday were derived.

4. Data Cleaning and Preprocessing :

Performed Using Python (Pandas)

Tasks Done:

- Checked missing values.
 - Identified duplicate records.
 - Validated data quality and field consistency.
 - Converted expiry_date and timestamp columns to datetime format.
 - Created derived columns from timestamp:
 - claim_date
 - claim_time
 - weekday
 - Prepared cleaned data for SQL analysis and Streamlit dashboard development.
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5. Python EDA (Exploratory Data Analysis) :

The following visualizations were created:

Univariate

1. Provider Type Distribution
2. Receiver Type Distribution
3. Food Type Distribution
4. Meal Type Distribution

Bivariate

5. City vs Food Listings
6. Provider Type vs Quantity
7. Food Type vs Quantity
8. Meal Type vs Quantity

Multivariate

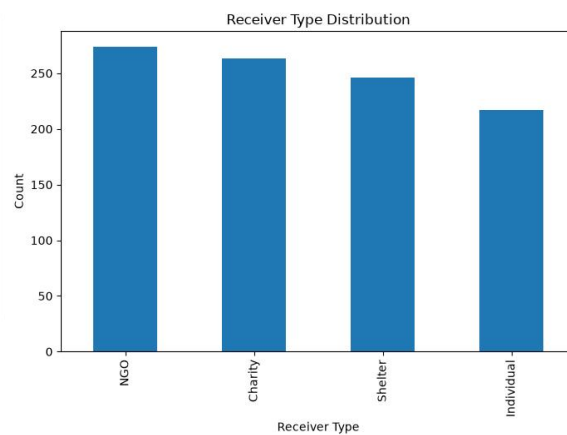
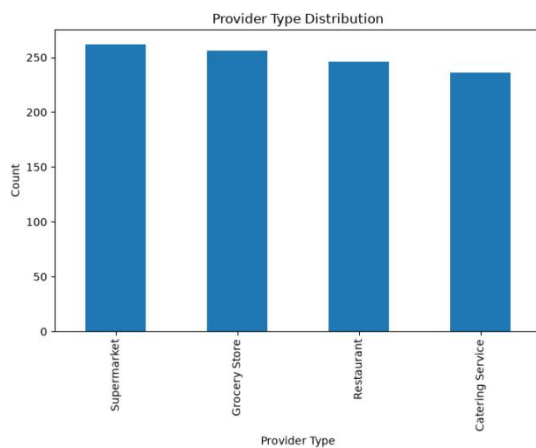
9. City + Provider Type + Quantity
10. Food Type + Meal Type + Quantity
11. Provider + Claims + Quantity
12. Receiver + Claims + Quantity

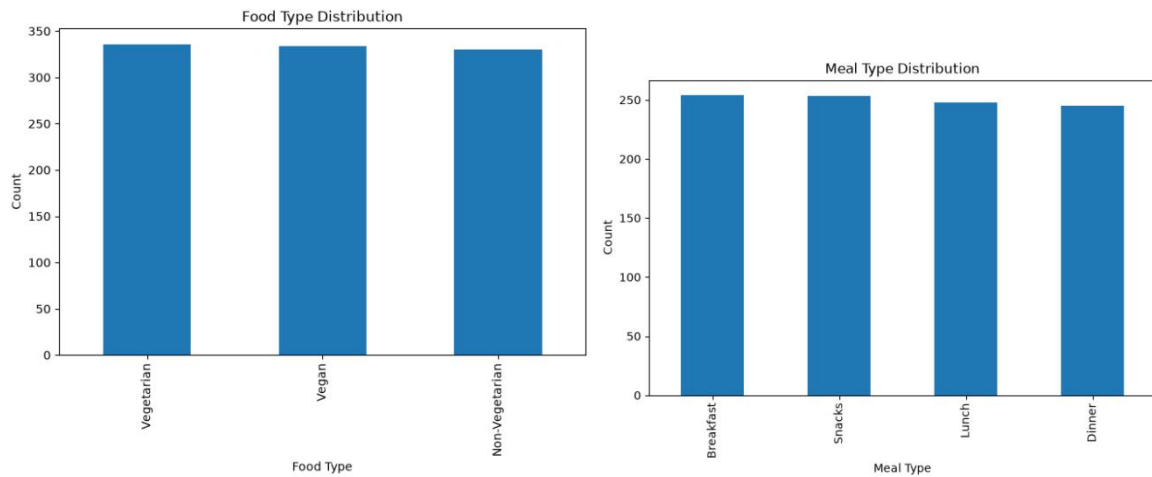
Claim Analysis

13. Claim Status Distribution
14. Top Receivers
15. Top Providers

Additional Analysis

16. Food Quantity Distribution
17. Claims by Weekday
18. Food Listing by Expiry date





5. SQL Analysis :

SQL queries were performed to answer business questions such as:

- How many food providers and receivers are there in each city?

Results 1

select coalesce(p.city,r.city) as city , p. providers_count , r. receivers_count

	AZ city	123 providers_count	123 receivers_count
1	Brandonhaven	[NULL]	2
2	Kelleystad	[NULL]	1
3	Lake Victoriaton	[NULL]	1
4	Riverafort	[NULL]	1
5	Rosaleschester	[NULL]	1
6	West Cherylland	[NULL]	1
7	Patriciamouth	[NULL]	1

- Which type of food provider contributes the most food?
- What is the contact information of food providers in a specific city?
- Which receivers have claimed the most food?
- What is the total quantity of food available from all providers?
- Which city has the highest number of food listings?
- What are the most commonly available food types?
- How many food claims have been made for each food item?
- Which provider has had the highest number of successful food claims?
- What percentage of food claims are completed vs pending vs cancelled?
- What is the average quantity of food claimed per receiver?
- Which meal type is claimed the most?
- What is the total quantity donated by each provider?

- Which cities have more providers than receivers?
 - Which cities have the highest amount of unclaimed food?
 - Which providers have the highest claim success rates?
 - Which food type has the highest cancellation rate?
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6. Streamlit Dashboard :

Link:(<https://foodwastemanagement-avmmbnbyx68vpf7hsgkq6d.streamlit.app/>)

Dashboard Page

- Summary metrics

EDA Page

- Visualizations
- Data exploration

SQL Analysis Page

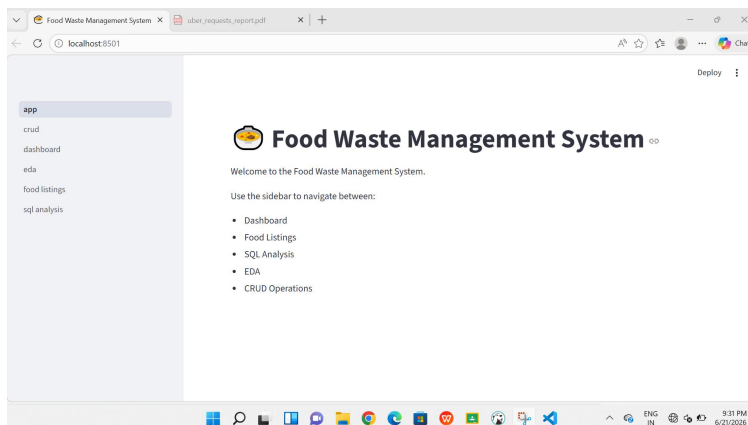
- Business query execution
- Results interpretation

CRUD Page

- Explain Create, Read, Update and Delete functionality for:
 - Providers
 - Receivers
 - Food Listings

Food Listings Page

- Listing management
- Data display
- Filtering



7. Key Insights :

1. Food Listings Concentrated in Specific Cities
 2. Certain Provider Types Donate More Food
 3. Food Availability Exceeds Demand in Some Cities
 4. Popular Food Types Receive More Claims
 5. Claim Completion Rate Reflects Platform Efficiency
 6. Some Providers Have Higher Claim Success
 7. Unclaimed Food Exists in Multiple Cities
 8. Meal Type Preferences are Visible
 9. Food Claim Activity Varies by Weekday
 10. Cancellation Patterns Differ Across Food Types
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8. Business Recommendations :

1. Increase Receiver Registration in Low-Demand Cities
2. Promote High-Performing Providers
3. Reduce Food Wastage Through Better Matching
4. Prioritize Popular Food Categories
5. Improve Notification Systems
6. Monitor Unclaimed Food Listings
7. Investigate Claim Cancellations
8. Expand Provider Network
9. Use Weekly Demand Patterns
10. Develop a Real-Time Dashboard